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Iterative Methods for Solving A System of Linear Equations

Chapter Learning Outcomes

CLO

4

- Know different iterative methods for solving a system of different linear equations.
- Know and understand the sufficient condition to start Jacobi method.
- Apply Jacobi method to solve a system of different linear equations.
- Know and understand the sufficient condition to start Gauss-Seidel method.
- Apply Gauss-Seidel method to solve a system of different linear equations.

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Solving a System of Linear Equations

Direct Methods

1- Cramer's Rule

2- Gauss Elimination

3- Gauss-Jordan Elimination

4- Inverse Rule

Iterative Methods

1- Jacobi Method

2- Gauss-Seidel Method

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Jacobi Method

Jacobi Method

The following is a system of linear equations:

$$a_1 x + b_1 y + c_1 z = d_1$$
$$a_2 x + b_2 y + c_2 z = d_2$$
$$a_3 x + b_3 y + c_3 z = d_3$$

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Jacobi Method

For the following equation, try to find the value of ($\boxed{x}$).

$$a_1 x + b_1 y + c_1 z = d_1$$

$$x = \frac{d_1 - b_1 y - c_1 z}{a_1}$$

Jacobi Method

For the following equation, try to find the value of ($\boxed{y}$).

$$a_2x + b_2y + c_2z = d_2$$

$$y = \frac{d_2 - a_2x - c_2z}{b_2}$$

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Jacobi Method

For the following equation, try to find the value of ($\boxed{z}$).

$$a_3x + b_3y + c_3z = d_3$$

$$z = \frac{d_3 - a_3x - b_3y}{c_3}$$

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Jacobi Method

Sufficient Condition to Start Jacobi Method

Row-Pivot Dominant

$$a_1x + b_1y + c_1z = d_1 \qquad |a_1| \geq |b_1| + |c_1|$$

$$a_2x + b_2y + c_2z = d_2 \qquad |b_2| \geq |a_2| + |c_2|$$

$$a_3x + b_3y + c_3z = d_3 \qquad |c_3| \geq |a_3| + |b_3|$$

Jacobi Method

EXAMPLE

Solve the following system of linear equations using Jacobi Method. Assume the tolerance is 10^{-3} and the initial guess is (0, 0, 0):

$$5x - 2y + 3z = -1$$
$$-3x + 9y + z = 2$$
$$2x - y - 7z = 3$$

Jacobi Method

Step 1 : Check the sufficient condition (row-pivot dominant) to start Jacobi method.

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Jacobi Method

Step 2 : Write the general formulas for Jacobi method.

$$5x - 2y + 3z = -1$$

$$x^k = \frac{-1 + 2y^{k-1} - 3z^{k-1}}{5}$$

$$-3x + 9y + z = 2$$

$$y^k = \frac{2 + 3x^{k-1} - z^{k-1}}{9}$$

$$2x - y - 7z = 3$$

$$z^k = \frac{3 - 2x^{k-1} + y^{k-1}}{-7}$$

Jacobi Method

Step 3 : Assume the initial guess: $x^0 = 0,\ y^0 = 0,\ z^0 = 0$

Step 4 : (Iteration 1) Find the values $x^1,\ y^1,\ z^1$

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Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1}\rvert$	$\lvert y^k - y^{k-1}\rvert$	$\lvert z^k - z^{k-1}\rvert$	Stop?
0	0	0	0	----	----	----	No
1							No

When to Stop ?

If $\lvert x^k - x^{k-1}\rvert \le \varepsilon$ and $\lvert y^k - y^{k-1}\rvert \le \varepsilon$ and $\lvert z^k - z^{k-1}\rvert \le \varepsilon$.

Jacobi Method

Step 5 : (Iteration 2) Find the values x^2, y^2, z^2

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Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							

Jacobi Method

Step 6 : (Iteration 3) Find the values x^3, y^3, z^3

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Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							

Jacobi Method

Step 7 : (Iteration 3) Find the values x^4, y^4, z^4

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Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							

Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							
5							

Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							
5							
6							

Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							
5							
6							
7							

Jacobi Method

The solution of the following system is:

$$5x - 2y + 3z = -1$$
$$-3x + 9y + z = 2$$
$$2x - y - 7z = 3$$

x = 0.1861 y = 0.3313 z = -0.4226

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Jacobi Method

Check your answer:

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Jacobi Method – Example 2

EXAMPLE

Solve the following system of linear equations using Jacobi Method. Assume the tolerance is 10^{-3} and the initial guess is (0, 0, 0):

$$\begin{aligned} -3x + 22y + 2z &= 47 \\ 5x + y + 20z &= 67 \\ 45x + 2y + 3z &= 58 \end{aligned}$$

Jacobi Method – Example 2

Step 1 : Check the sufficient condition (row-pivot dominant) to start Jacobi method.

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Jacobi Method – Example 2

Step 2 : Try to make a swap between equations.

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Jacobi Method – Example 2

Step 3 : Write the general formulas for Jacobi method.

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Jacobi Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							
5							
6							

Jacobi Method – Example 2

The solution for the system is listed below:

$$-3x + 22y + 2z = 47$$
$$5x + y + 20z = 67$$
$$45x + 2y + 3z = 58$$

x = 1 | y = 2 | z = 3

Solving a System of Linear Equations

Direct Methods

- ✓ 1- Cramer's Rule
- ✓ 2- Gauss Elimination
- ✓ 3- Gauss-Jordan Elimination
- ✓ 4- Inverse Rule

Iterative Methods

- ✓ 1- Jacobi Method
- 2- Gauss-Seidel Method

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Gauss-Seidel Method

Gauss-Seidel Method

The following is a system of linear equation:

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

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Gauss-Seidel Method

For the following equation, try to find the value of ($\boxed{x}$).

$$a_1 x + b_1 y + c_1 z = d_1$$

$$x = \frac{d_1 - b_1 y - c_1 z}{a_1}$$

Gauss-Seidel Method

For the following equation, try to find the value of ($\boxed{y}$).

$$a_2x + b_2y + c_2z = d_2$$

$$y = \frac{d_2 - a_2x - c_2z}{b_2}$$

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Gauss-Seidel Method

For the following equation, try to find the value of ($\boxed{z}$).

$$a_3x + b_3y + c_3z = d_3$$

$$z = \frac{d_3 - a_3x - b_3y}{c_3}$$

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Gauss-Seidel Method

Sufficient Condition to Start Gauess-Seidel Method

Row-Pivot Dominant

$$a_1x + b_1y + c_1z = d_1 \qquad |a_1| \geq |b_1| + |c_1|$$

$$a_2x + b_2y + c_2z = d_2 \qquad |b_2| \geq |a_2| + |c_2|$$

$$a_3x + b_3y + c_3z = d_3 \qquad |c_3| \geq |a_3| + |b_3|$$

Gauss-Seidel– Example

EXAMPLE

Solve the following system of linear equations using **Gauss-Seidel Method**. Assume the tolerance is 10^{-3} and the initial guess is (0, 0, 0):

$$5x - 2y + 3z = -1$$
$$-3x + 9y + z = 2$$
$$2x - y - 7z = 3$$

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Compare Gauss-Seidel and Jacobi

$$
\begin{aligned}
5x - 2y + 3z &= -1 \\
-3x + 9y + z &= 2 \\
2x - y - 7z &= 3
\end{aligned}
$$

Jacobi

Gauss-Seidel

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Gauss-Seidel Method

Step 1 : Check the sufficient condition to start Gauss-Seidel method.

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Gauss-Seidel Method

Step 2 : Write the general formulas for Gauss-Seidel method.

$$5x - 2y + 3z = -1$$

$$x^k = \frac{-1 + 2y^{k-1} - 3z^{k-1}}{5}$$

$$-3x + 9y + z = 2$$

$$y^k = \frac{2 + 3x^k - z^{k-1}}{9}$$

$$2x - y - 7z = 3$$

$$z^k = \frac{3 - 2x^k + y^k}{-7}$$

Gauss-Seidel Method

Step 3 : Assume the initial guess: $x^0 = 0,\ y^0 = 0,\ z^0 = 0$

Step 4 : (Iteration 1) Find the values $x^1,\ y^1,\ z^1$

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Gauss-Seidel Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1}\rvert$	$\lvert y^k - y^{k-1}\rvert$	$\lvert z^k - z^{k-1}\rvert$	Stop?
0	0	0	0	----	----	----	No
1							No

When to Stop ?

If $|x^k - x^{k-1}| \le \varepsilon$ and $|y^k - y^{k-1}| \le \varepsilon$ and $|z^k - z^{k-1}| \le \varepsilon$.

Gauss-Seidel Method

Step 5 : (Iteration 2) Find the values x^2, y^2, z^2

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Gauss-Seidel Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							

Gauss-Seidel Method

Step 6 : (Iteration 3) Find the values x^3, y^3, z^3

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Gauss-Seidel Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							

Gauss-Seidel Method

Step 7 : (Iteration 3) Find the values x^4, y^4, z^4

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Gauss-Seidel Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							

Gauss-Seidel Method

Step 3 : (Iteration 5) Find the values x^5, y^5, z^5

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Gauss-Seidel Method

k	x^k	y^k	z^k	$\lvert x^k - x^{k-1} \rvert$	$\lvert y^k - y^{k-1} \rvert$	$\lvert z^k - z^{k-1} \rvert$	Stop?
0	0	0	0	----	----	----	
1							
2							
3							
4							
5							

Gauss-Seidel Method

The solution of the following system is:

$$5x - 2y + 3z = -1$$
$$-3x + 9y + z = 2$$
$$2x - y - 7z = 3$$

x = 0.1860 | y = 0.3312 | z = -0.4227

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Gauss-Seidel Method

Check your answer:

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Check - Chapter Learning Outcomes

CLO

4

- Know different iterative methods for solving a system of different linear equations.
- Know and understand the sufficient condition to start Jacobi method.
- Apply Jacobi method to solve a system of different linear equations.
- Know and understand the sufficient condition to start Gauss-Seidel method.
- Apply Gauss-Seidel method to solve a system of different linear equations.

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End of Chapter

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